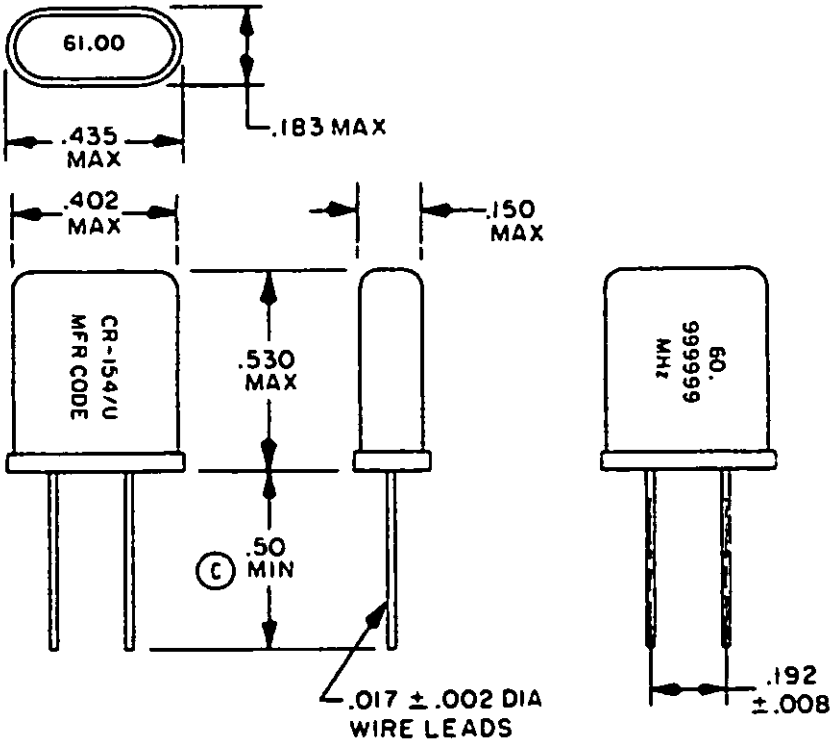
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-154/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 17 to 61 MHz; third overtone; controlled; series.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-154/U.

© denotes changes

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© TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure 1.
Frequency range	17 to 61 MHz, inclusive
Capacitance, shunt	4.0 pF, maximum
Mode of oscillation	Third mechanical overtone
Test set, calibration values and rated drive level	See table II
Reference temperature	85°C ±1°C
Operating temperature range (controlled)	80°C to 90°C, inclusive
Frequency tolerance	±20 ppm
Equivalent resistance	40 ohms, maximum
Frequency stability	±5 ppm
Operable temperature range	-55°C to +180°C, inclusive
Unwanted modes	Method II, minimum unwanted-to-main mode resistance ratio of 2:1 or 40 ohms, whichever is greater
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 ppm

1/ Method 204 of MIL-STD-202, test condition A.

TABLE II. Test, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values		Rated drive level
		Resistance	Resistor voltage drop	
	MHz	Ohms	Volts	mW
TS-683/TSM	17 to 25	40	0.28	2.0 ±0.4
	25 + to 61	40	0.20	1.0 ±0.2

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

User activities:
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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